PHYSICS

SECTION – I
Single Correct Choice Type

This section contains 4 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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1. A small, hollow metal cylinder is closed at one end. Its mass is known which of the following instruments are required to find the density of the metal?

(a) slide callipers only (b) slide callipers and screw gauge

(c) screw gauge and spherometer (d) slide callipers and spherometer

2. Protons accelerated through a potential difference V travels in the form of beams of radius r and constitutes a current *I*. The electric field at the surface of the beams is (take e as charge of proton and mass of the proton is m)

(a) $\frac{I}{2\pi r\varepsilon_0}\sqrt{\frac{m}{2\,\text{eV}}}$ (b) $\frac{I}{4\pi r\varepsilon_0}\sqrt{\frac{m}{2\,\text{eV}}}$ (c) $\frac{I}{2\pi r\varepsilon_0}\sqrt{\frac{m}{\text{eV}}}$ (d) $\frac{I}{2\pi r\varepsilon_0}\sqrt{\frac{m}{4\,\text{eV}}}$

3. Let us assume that the temperature of the air varies with height linearly from the value T_1 = 300 K at y_1 = 0 to T_2 = 250 K at y_2 = 10 km. The time *t* needed for a sound wave produced at the earth's surface to reach the height y_2 is (velocity of sound at 0^0C = 330 m/s)

(a) 30 s (b) 60 s (c) 15 s (d) 90 s

4. The internal energy of an imaginary gas is determined by the equation

$$U = a\ln\left(\frac{T}{T_0}\right) + b\ln\left(\frac{P}{P_0}\right)$$

Where a = 3 kJ, b = 7 kJ, T_0 = 200 K and P_0 = 10 KPa

On importing 500 J of heat of constant pressure of P = 10^5 Pa the volume changes (increase) by 0.5×10^{-3} m^3. If final temperature is *n* times the initial temperature then *n* is

(a) 1.00 (b) 1.17 (c) 2.00 (d) 2.17

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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5. A capacitor of capacity C has charge q_0 in fully charged condition. The capacitor is now connected to a coil of inductance L. Which of following are correct?

(a) at an instant $t = T/8$, charge $= \frac{q_0}{\sqrt{2}}$

(b) magnetic energy at $t = \frac{T}{8}$ is 50% of the total energy

(c) ratio of magnetic energy to that of electorate energy is $\frac{1}{2}$ at any instant

(d) ratio of electrostatic energy to magnetic energy is 1 only at q is 70.7 % of maximum charge.

6. A sphere of radius R uniformly charged with surface charge density σ, rotates around the axis passing through its centre at an angular velocity ω. Now

(a) magnetic field at centre $= \frac{2}{3}\mu_0\omega\sigma R$

(b) magnetic moment $= \frac{4}{3}\pi\sigma R^4\omega$

(c) magnetic field at centre $= \frac{1}{3}\mu_0\omega\sigma R$

(d) magnetic moment $= \frac{2}{3}\pi\sigma R^4\omega$

7. A sphere of mass m = 200 gm fixed between 2 springs of total force constant k = 28.8 N/m can slide over a frictionless platform PQ placed horizontally. The system rotates with an angular velocity ω = 8 rad/sec about a vertical axis OO'. Now,

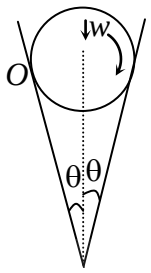
(a) time period of system is 0.7 sec and

(b) time period of system is 0.4 sec and for ω = 12 rad/sec, it stops

(c) time period of system is 0.7 sec and for ω = 12 rad/sec, the oscillator will stop

(d) time period of system is 0.4 sec and for all value of ω, the oscillation continues

8. A rotor of weight w and radius r rests in a v-shaped support as shown. The coefficient of friction at the contact is μ. The couple required to produce rotation is



(a) $\frac{\mu wr}{(1+\mu^2)\sin\theta}$ maintaining the contact O and wr cos θ in the absence of contact O

(b) $\frac{\mu wr}{(1+\mu^2)\sin\theta}$ when in contact at O is maintained

(c) wr cos θ in the absence of contact O

(d) $\frac{2\mu wr}{(1+\mu^2)\cos\theta}$ maintaining the contact O and wr cos θ in the absence of contact O.

SECTION – III
Reasoning Type

This section contains 4 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

9. **Statement-1:** The rate of change of momentum of a many particle system is proportional to the vector sum of the internal forces.

Statement-2: Internal forces can change the kinetic energy as well as momentum of the body.

(a) A (b) B (c) C (d) D

10. **Statement-1:** If the sum of the 2 unit vectors is also a unit vector, then magnitude of their difference is square root of three.

Statement-2: To find resultant of 2 vectors, we use triangle law.

(a) A (b) B (c) C (d) D

11. **Statement-1:** The images formed by total internal reflections are much brighter than those formed by mirrors or lens.

Statement-2: There is no lose of intensity in total internal reflections.

(a) A (b) B (c) C (d) D

12. **Statement-1:** If the frequency of the light incident on a metal surface is halved, the maximum K.E. of emitted photoelectrons also gets halved.

Statement-2: K.E. of a particle is directly proportional to frequency.

(a) A (b) B (c) C (d) D

SECTION–IV
Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage-I

A resistor R and an inductor L are connected in series to a battery of volt (V) with zero internal resistance. Due to self inductor an induced emf is setup across L. Magnetic energy is stored in an inductor. Use R = 5Ω L = 500 H V = 10 V

13. What is the current in the circuit when the rate of increase of energy in inductor is maximum

(a) 1 amp (b) 2 amp (c) 3 amp (d) 4 amp

14. What is the maximum current in the circuit?

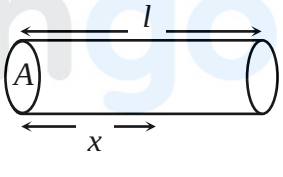
(a) 1 amp (b) 2 amp (c) 3 amp (d) 4 amp

15. When will the joule heat dissipated in the resistor change at the fastest rate?

(a) 69 sec (b) 34 sec (c) 138 sec (d) 40 sec

Passage-II

The cross sectional area and length of a cylindrical conductor are A and l respectively. The specified conductivity varies as $\sigma = \sigma_0 \frac{l}{\sqrt{x}}$, when x is the distance along the axis of the cylinder from one of its ends.



16. What is the resistance of cylinder along its cylindrical axis?

(a) $\frac{3\sqrt{l}}{2A\sigma_0}$ (b) $\frac{\sqrt{l}}{3A\sigma_0}$ (c) $\frac{2\sqrt{l}}{A\sigma_0}$ (d) $\frac{2\sqrt{l}}{3A\sigma_0}$

17. What is the current density it potential does along the cylinder v_0?

(a) $\frac{3\sigma_0 v_0}{2\sqrt{l}}$ (b) $\frac{2\sigma_0 v_0}{3\sqrt{l}}$ (c) $\frac{\sigma_0 v_0}{2\sqrt{l}}$ (d) $\frac{3\sigma_0 v_0}{\sqrt{l}}$

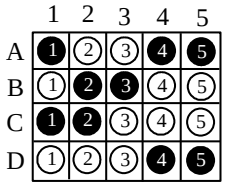
18. The electric field at each point in the cylinder in this case is

(a) $\frac{3v_0\sqrt{x}}{2(l)^{3/2}}$ (b) $\frac{2v_0\sqrt{x}}{3(l)^{3/2}}$ (c) $\frac{v_0\sqrt{x}}{(l)^{3/2}}$ (d) $\frac{2v_0\sqrt{x}}{(l)^{3/2}}$

SECTION–V
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

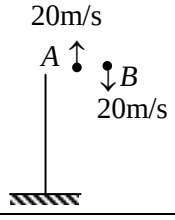
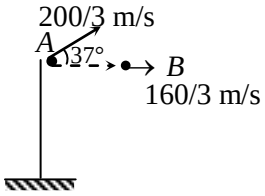
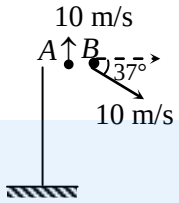
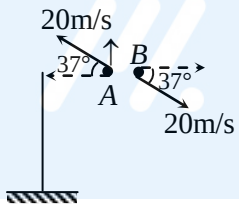


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1 Some situations are given with reference to Young's double slit experiment in column I and their effects are mentioned in column II. Match these situations with their corresponding effects.

Column I		Column II	
(A)	Separation between screen and slits is increased	**1.**	fringe width increases
(B)	Slit separation is increases symmetrically	**2.**	fringe width decreases
(C)	The whole arrangement is immersed in water	**3.**	central maxima shifts either up or down
(D)	A thin slab is introduced in front of any one of the source	**4.**	all the fringes shifts by the same distance
		5.	all the fringes shifts by different distances

2. Both *A* and *B* are thrown simultaneously as shown from a high tower. Match column - I with Column - II

Column - I	**Column -II**
(A) 	**1.** Distance between the two balls after 2 seconds is $16\sqrt{5}$ m.
(B) 	**2.** Distance between the two balls after 2 seconds is 80 m.
(C) 	**3.** Magnitude of relative velocity of *B* with respect to *A* is $8\sqrt{5}$ m/s.
(D) 	**4.** Magnitude of relative velocity of *B* with respect to *A* is 40 m/s. **5.** Magnitude of relative horizontal separation of A with respect to B is zero.